A 40.291 day transit candidate on KIC 6146676

Vikram Pandya[1]

[1] *SP Jain School of Global Management, Mumbai, India*

ABSTRACT

This note reports a transit candidate on KIC 6146676, a star Kepler recorded only in other targets' pixels. Its KBonus light curve shows a 3.69 hour, 2,214 ppm dip every 40.29081 days over 33 transits. In difference images the dip is nearer this star than its brightest neighbour in 4 of 4 observing seasons, 2 of them with a box cut by the stamp edge. The planet would be about 2.5 Earth radii. Null searches give a false-alarm rate of 1 in 170 per star, about 9 such peaks expected across the 1,579-star sweep; TRICERATOPS 1.1.0 gives a false-positive probability of 0.073, or 0.055 with the 6 neighbours the difference images exclude removed (no validation class applies at 30-minute cadence; Giacalone et al. 2021); a four-scenario estimate without a bound-companion or diluted-planet scenario gives 0.0033, a lower bound. It remains a candidate pending follow-up.

Keywords: Exoplanet detection methods (489), Exoplanets (498), Transit photometry (1709)

1. THE SIGNAL

In the KBonus Kepler Background light curve (Martínez-Palomera et al. 2023a) of KIC 6146676 (Gaia DR3 2073896294004902272, $G = 16.2$), recorded by Kepler (Borucki et al. 2010) only in other targets' pixels, a dip of 2,214 ± 114 ppm lasting $3.686^{+0.063}_{-0.072}$ hours repeats every 40.29081 ± 0.00015 days, with the reference mid-transit at $\mathrm{BJD_{TDB}}$ 2455734.3205 ± 0.0014 at epoch 19, the centre of the series. Timed singly, 31 of 32 transits fit this linear ephemeris with an rms of 9.4 minutes; the depth and shape below use the linear-fit ephemeris.

2. DETECTION AND VETTING

SkySift, which vets each signal through a fixed ladder of checks, found the signal by box search over 0.5 to 200 days periods: signal detection efficiency 50.5, signal-to-noise ratio 30.0, 33 transits. 32 of 37 predicted transits with data are well covered; the largest single-event statistic is 0.28 of the multiple-event statistic. Fitting a quadratic limb-darkened model (Claret & Bloemen 2011) on a circular orbit, integrated over the 29.4-minute exposure, gives a radius ratio of $0.0454^{+0.0011}_{-0.0012}$; b and a/R* are unconstrained at this cadence (the quoted values follow the host's density): b is $0.457^{+0.054}_{-0.061}$ and a/R* $78.4^{+2.5}_{-2.3}$. The odd and even depths are 2,276 ± 177 and 2,156 ± 151 ppm per event (0.5 sigma apart). The phase 0.5 depth is 95 ± 128 ppm (3-sigma limit 480 ppm), the strongest dip elsewhere 342 ppm at phase 0.38 (3.8 sigma); positive excursions in the same fold reach 5.2 sigma. The depth agrees across 4 quarter groups (quarters modulo 4; chi-squared 0.4 for 3 degrees of freedom, p = 0.94), though the per-event depth deepens by 228 ± 90 ppm per year (2.5 sigma, scaled errors). It matches no Kepler KOI, DR25 TCE, TESS TOI or Kepler eclipsing binary entry.

3. WHICH STAR IS DIMMING

In difference images the centroid sits a median 0.19 pixels from this star and 2.82 from its brightest neighbour, nearer this star in 4 of 4 observing seasons, its error carrying a pointing-jitter floor of 0.017 px. In seasons 0 and 3 this star's 3 x 3 box is cut by the stamp edge, adding up to 0.08 px to the centroid error. Against the Gaia position the centroid sits 0.25 ± 0.28 arcsec away, with an exclusion radius of 0.83 arcsec (max(3 sigma, 0.5 arcsec)) within which a background source is not excluded; the nearest Gaia source beyond it lies at 6.9 arcsec (G = 19.8); against the star's out-of-transit image the offset is 1.8 ± 1.0 arcsec (1.8 sigma), not an offset by the rule of Bryson et al. (2013) (over 3 sigma and 0.2 arcsec), its reference pulled by the neighbour up to 1.30 px. A two-source fit gives a median depth of 2,353 ppm on this star and -21 on the neighbour. The depth is KBonus PSF photometry, decontaminated to first order, with no crowding correction applied; the SAP depth is 0.696 ± 0.069 of it, 2.3 sigma short of a CROWDSAP

(this star's light fraction in the aperture) of 0.86, with a PSF flux fraction inside the aperture of 0.80. The target's light predicts 4 ppm in KIC 6146666's optimal aperture against -9 ± 12 ppm measured.

4. THE STAR AND PLANET

No Kepler stellar catalogue covers a background star, so the parameters come from Gaia DR3 (Gaia Collaboration et al. 2016, 2023) colour and parallax; Gaia gives a RUWE of 0.94 and no non-single-star solution. The reddening is E(B-V) = 0.009 (0.009 to 0.014) from Bayestar19 (Green et al. 2019) at 303 pc, nearer than the map's reliable range (from 405 pc; full column 0.674 (Schlafly & Finkbeiner 2011), an upper bound). The colour gives a spectral type of M1V and an effective temperature of $3{,}666^{+90}_{-78}$ K, and the zero-point-corrected parallax a distance of $303.0^{+3.6}_{-3.5}$ pc. From the absolute K_s magnitude the relations of Mann et al. (2015) and Mann et al. (2019) give a radius of $0.500^{+0.015}_{-0.016}$ $R_\odot$ and a mass of 0.498 ± 0.012 $M_\odot$. With the radius ratio of 0.0454 the planet radius is 2.5 ± 0.1 $R_\oplus$, on an orbit of 0.1823 ± 0.0015 AU with an equilibrium temperature of 268^{+8}_{-7} K at a Bond albedo of 0.3 and $1.22^{+0.14}_{-0.12}$ times Earth's insolation.

5. HOW LIKELY IS A FALSE POSITIVE

11 of 1,998 null searches of other KBonus stars reached the signal's SDE (1 in 170); across the 1,579-star sweep about 9 such peaks are expected by chance, and the vetting ladder, not the SDE, separates the candidate from them. TRICERATOPS 1.1.0 (Giacalone et al. 2021), run on the KBonus light curve with Kepler limb darkening and Gaia DR3 field stars, gives a median false-positive probability of 0.073 (0.022 without a planet transiting an unresolved bound companion, STP, absent from SkySift's model; 0.055, and 1 in 3,000 without STP, with the 6 neighbours the difference images exclude removed) and a nearby false-positive probability of 0.020 over 20 runs (no validation class applies at 30-minute cadence). The pixel-excluded value (0.055, mainly STP) rests on our own evidence, the difference-image centroids; the all-neighbours value (0.073) rests on neighbour shares set by the PRF overlap rule within the catalogue's crowding budget, a bound rather than a measurement. A four-scenario estimate without a bound-companion or diluted-planet scenario gives 0.0033, a lower bound. TESS observed the star in 9 sectors; the predicted folded significance of 0.54 to 0.88 sigma means no detection is expected at the nominal noise; no TESS data were examined (Christ et al. 2019).

6. WHAT IS NOT YET KNOWN

No spectrum: the planet radius depends on the stellar radius from Gaia colour and parallax.

7. DATA AVAILABILITY

Pack (data and methods): doi:10.5281/zenodo.23684431; light curves: the KBonus Kepler Background high-level science product at MAST (Martínez-Palomera et al. 2023b); code: SkySift at commit f542d65, doi:10.5281/zenodo.23843666.

Software: SkySift 0.1.0 (search version 5+6cfe4cf4, commit f542d65), numpy 2.5.4, scipy 1.18.1, astropy 8.0.1, matplotlib 3.11.2, triceratops 1.1.0

Facilities: Kepler, Gaia, MAST, Exoplanet Archive

REFERENCES

Borucki, W. J., Koch, D., Basri, G., et al. 2010, Science, 327, 977

Bryson, S. T., Jenkins, J. M., Gilliland, R. L., et al. 2013, PASP, 125, 889

Christ, C. N., Montet, B. T., & Fabrycky, D. C. 2019, AJ, 157, 235

Claret, A., & Bloemen, S. 2011, A&A, 529, A75

Gaia Collaboration, Prusti, T., de Bruijne, J. H. J., et al. 2016, A&A, 595, A1

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, A&A, 674, A1

Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, AJ, 161, 24

Green, G. M., Schlafly, E., Zucker, C., Speagle, J. S., & Finkbeiner, D. 2019, ApJ, 887, 93

Martínez-Palomera, J., Hedges, C., & Dotson, J. 2023a, AJ, 166, 265, doi:10.3847/1538-3881/ad0727

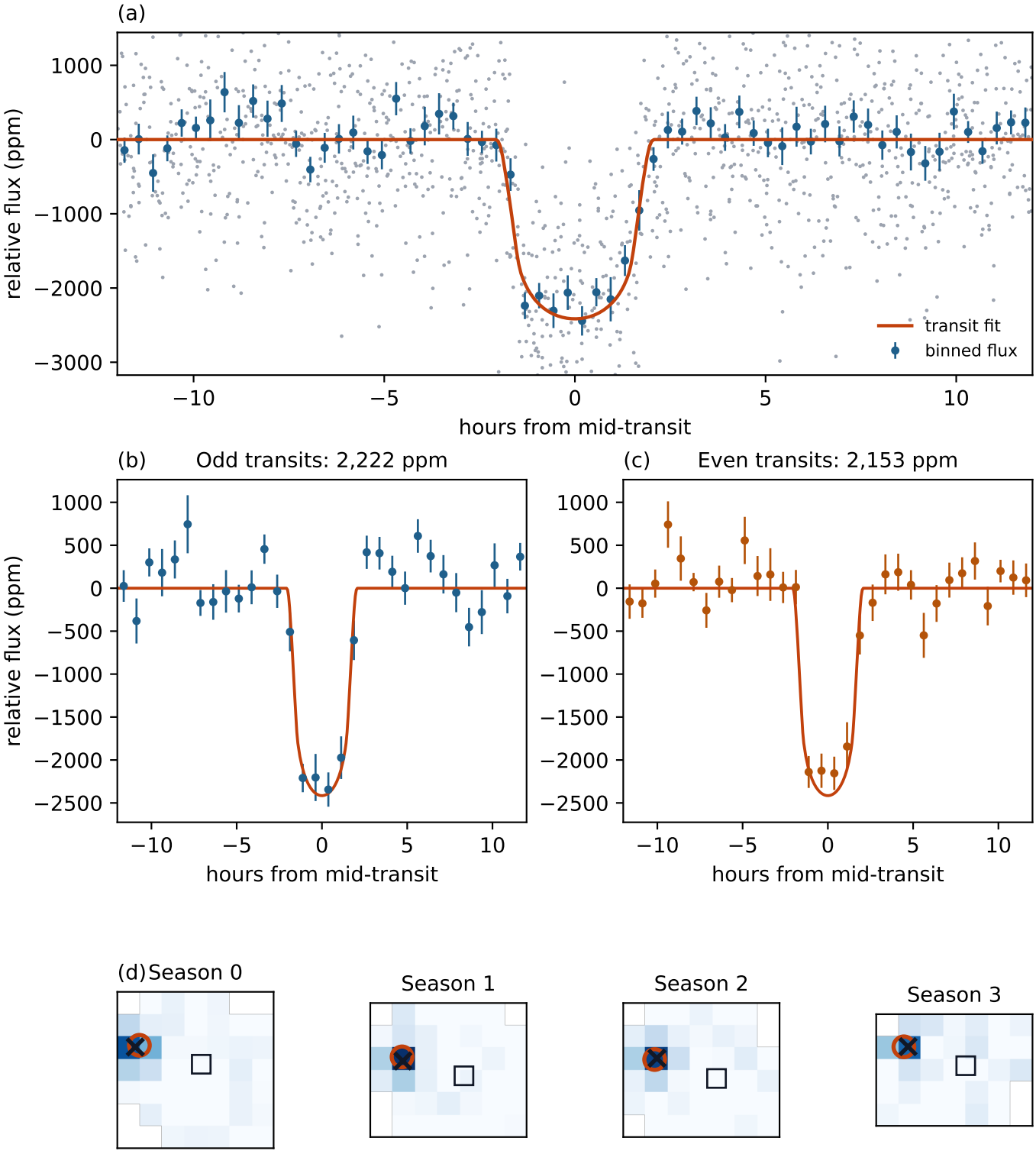


Figure 1. (a) the light curve folded on the 40.29081 day linear-fit ephemeris, within 12.0 hours of mid-transit, in parts per million: bins of 22.5 minutes with standard errors (dots) and the transit fit (line); (b) odd and (c) even transits on the same scale; (d) per-season difference images: this star (circle), the neighbour (square), the centroid (cross).

Martínez-Palomera, J., Hedges, C., & Dotson, J. 2023b, KBonus Kepler Background (KBONUS-BKG) high-level science product, MAST, doi:10.17909/7jbr-w430

Mann, A. W., Feiden, G. A., Gaidos, E., et al. 2015, ApJ, 804, 64

Mann, A. W., Dupuy, T., Kraus, A. L., et al. 2019, ApJ, 871, 63

Schlafly, E. F., & Finkbeiner, D. P. 2011, ApJ, 737, 103